Secure IPsec Tunnel Demo Execution Guide

Objective

To execute and verify a secure IPsec tunnel between the Central Unit (CU) and Distributed Unit (DU) using a containerized setup.

Prerequisites

- Docker and Docker Compose must be installed and running.
- Project directory (~/crypto) must be present with proper configuration files.
- Containers oai-cu-pqc and oai-du-pqc must be defined in docker-compose.yml.
- Network configuration (IP addressing and bridge network) must be correctly set.
- Optional: Wireshark installed for packet analysis.

Procedure

Step 0 : Navigate to Project Directory

```
cd ~/crypto
```

This command moves to the project directory where the configuration and Docker files are located.

Step 1 : Start Containers

```
docker-compose up -d
```

This command starts all defined containers in detached mode.

Step 2 : Verify Running Containers

```
docker ps
```

This command lists all active containers. Ensure that oai-cu-pqc and oai-du-pqc are running.

Step 3 : Start IPsec Daemon

```
docker exec -d oai-cu-pqc /usr/libexec/ipsec/charon

docker exec -d oai-du-pqc /usr/libexec/ipsec/charon
```

These commands start the StrongSwan IPsec daemon (charon) inside both CU and DU containers.

Step 4 : Load IPsec Configuration

```
docker exec -it oai-cu-pqc swanctl --load-all

docker exec -it oai-du-pqc swanctl --load-all
```

These commands load all IPsec configuration files into the daemon.
Expected result: configuration named f1-pqc is loaded.

Step 5 : Initiate Secure Tunnel

```
docker exec -it oai-cu-pqc swanctl --initiate --child f1-c
```

This command initiates the IPsec tunnel from the CU side using the defined child security association.
If duplicate CHILD_SA appears, it indicates the tunnel is already established.

Step 6 : Verify Tunnel Establishment

```
docker exec -it oai-cu-pqc swanctl --list-sas
```

This command displays active security associations.
Successful output must include:

- IKE_SA f1-pqc
- CHILD_SA f1-c

Step 7 : Generate Traffic

```
docker exec -it oai-cu-pqc ping 192.168.10.20
```

This command sends ICMP packets from CU to DU.
Successful replies confirm communication through the IPsec tunnel.

Step 8 — Capture Network Traffic (Optional)

Live Capture

```
sudo tcpdump -i br-2365720abad7
```

Captures live packets on the Docker bridge interface.

Save Capture to File

```
sudo tcpdump -i br-2365720abad7 -w capture.pcap
```

Stores captured packets in a file.

Analyze Capture

```
wireshark capture.pcap
```

Open the file in Wireshark and apply the filter:

```
icmp
```

This allows inspection of ICMP traffic flowing through the network.

Result

- Containers are running successfully.
- IPsec daemon is active in both CU and DU.
- Configuration is loaded correctly.
- Secure tunnel is established and verified.
- Data transmission occurs between CU and DU.
- Optional packet capture confirms network activity.